

Semester	III	L	T	P	C	COURSE CODE
Regulation	V23	0	0	3	1.5	V23CSL05
Name of the Course	Object Oriented Programming Through Java Lab					
Branch	Common to CSE, CST, CSE(AI) , AI&ML and CSE(DS)					

Syllabus Details

Course Outcomes:

After Successful completion of the Course, the student will be able to:

CO1:Construct programs to handle classes and objects. (K3)

CO2: Develop programs that incorporate inheritance and interfaces. (K3)

CO3:Construct programs on exception handling and File I/O. (K3)

CO4:Develop programs using multithreading and Java FX. (K3)

Experiments covering the Topics:

- Object Oriented Programming fundamentals- data types, control structures
- Classes, methods, objects, Inheritance, polymorphism,
- Exception handling, Threads, Packages, Interfaces
- Files, I/O streams, Java FX GUI

Exercise – 1:

- a) Develop a JAVA program to display default value of all primitive data type of JAVA
- b) Develop a java program that display the roots of a quadratic equation $ax^2+bx=0$. Find the discriminate D and basing on value of D, describe the nature of root.

Exercise – 2:

- a) Develop a JAVA program to search for an element in a given list of elements using binary search mechanism.
- b) Develop a JAVA program to sort for an element in a given list of elements using bubble sort
- c) Develop a JAVA program using String Buffer to delete, remove character.

Exercise – 3:

- a) Develop a JAVA program to implement class Mechanism. Build a class, methods and invoke them inside main method.
- b) Develop a JAVA program to implement method overloading.
- c) Develop a JAVA program to implement constructor.
- d) Develop a JAVA program to implement constructor overloading.

Exercise – 4:

- a) Develop a JAVA program to implement Single Inheritance.
- b) Develop a JAVA program to implement multi level Inheritance.
- c) Develop a JAVA program for abstract class to find areas of different shapes.

Exercise – 5:

- a) Develop a JAVA program on “super” keyword.
- b) Develop a JAVA program to implement Interface. What kind of Inheritance can be achieved?
- c) Develop a JAVA program that implements Runtime polymorphism.

Exercise – 6:

- a) Develop a JAVA program that describes exception handling mechanism.
- b) Develop a JAVA program on Multiple catch clauses.
- c) Develop a JAVA program to generate Built-in Exceptions.
- d) Develop a JAVA program to generate User Defined Exception.

Exercise – 7:

- a) Develop a JAVA program that creates threads by extending Thread class. First thread display “Good Morning “every 1 sec, the second thread displays “Hello “every 2 seconds and the third display “Welcome” every 3 seconds,(Repeat the same by implementing Runnable)
- b) Develop a program on isAlive() and join ().
- c) Demonstrate Daemon Threads.
- d) Demonstrate Producer Consumer Problem.

Exercise – 8:

- a) Develop a JAVA program that import and use the user defined packages.
- b) Demonstrate a GUI that display text in label and image in an ImageView (use JavaFX) Without writing any code.
- c) Develop a Tip Calculator app using several JavaFX components and learn how to respond to user interactions with the GUI

Textbooks:

1. JAVA one step ahead, Anitha Seth, B.L.Juneja, Oxford.
2. Joy with JAVA, Fundamentals of Object Oriented Programming, DebasisSamanta, MonalisaSarma, Cambridge, 2023.
3. JAVA 9 for Programmers, Paul Deitel, Harvey Deitel, 4th Edition, Pearson.

Reference Books:

1. The complete Reference Java, 11th edition, Herbert Schildt, TMH
2. Introduction to Java programming, 7th Edition, Y Daniel Liang, Pearson

Online Learning Resources:

1. <https://nptel.ac.in/courses/106/105/106105191/>
2. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_012880464547618816347_shared/overview